



Barker
College

2022

YEAR 12

ASSESSMENT TASK 3

Mathematics Advanced

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Student Number

Staff Involved:

- PCB* • AXD • GPF • ESP
- RAS • JZT • RJW • ALY

Wednesday 1st June, 8:30am

160 copies

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General Instructions

- Working time – 55 minutes
- Write using blue or black pen
- Write your Barker Student number in the spaces indicated
- Write your answers in the spaces provided
- Board approved calculators may be used
- The HSC Reference Sheet is provided.
- Attempt all questions
- ALL necessary working must be shown
- Marks may not be awarded for careless or badly arranged work
- Diagrams are NOT to scale

Topics Covered

- Exponential and Logarithmic Functions
- Trigonometric Functions

Section	Marks
Question 1	/ 10
Question 2	/ 10
Question 3	/ 10
Question 4	/ 10
Total Marks:	/ 40

Question 1 (10 marks)

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Student number

(a) Differentiate with respect to x :

(α) $y = \cos(4x)$ 1

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(β) $y = (2 + \ln x)^3$ 2

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(γ) $y = \frac{e^x}{x}$ 2

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(b) Find the following indefinite integrals:

(α) $\int \frac{1}{4x+1} dx$ 1

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(β) $\int x^3 e^{5x^4} dx$ 2

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(γ) $\int \frac{3x^2+x-2}{x} dx$ 2

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Student number

(i) Show that $\frac{dy}{dx} = e^{3x}(3x^2 - 13x + 4)$ **2**

(i) Show that $\frac{dy}{dx} = e^{3x}(3x^2 - 13x + 4)$

2

[illegible]

2

[illegible]

1

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(ii) Hence, find $\int \frac{(x+1)^2}{(x+1)^3+2} dx$

2

[illegible]

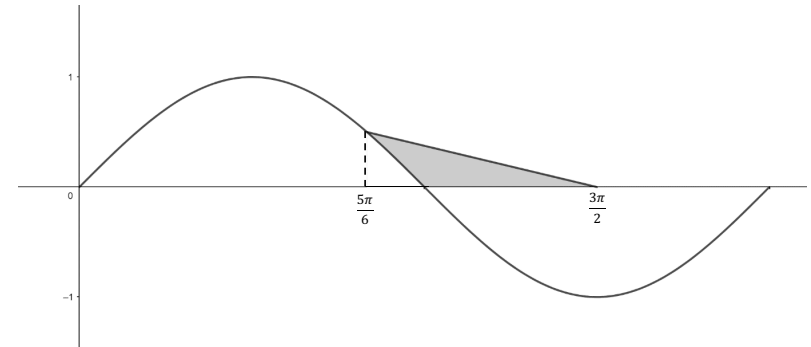
(iii) If $\int_0^{\sqrt[3]{22}-1} \frac{(x+1)^2}{(x+1)^3+2} dx = \ln b$, where b is an integer, find b .

3

[illegible]

- 4

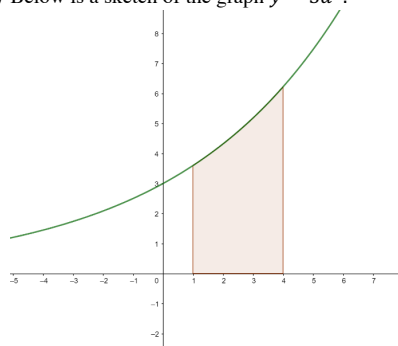
- 4

[illegible]

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Student number

(c) Below is a sketch of the graph $y = 3a^x$.



The value of $\int_1^4 3a^x dx = 14$.

Using graph transformations, find the value of $\int_2^5 3a^{x-1} - 3 dx$

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END OF ASSESSMENT

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END OF QUESTION 4

Student number

If you use this space, clearly indicate which question you are answering

[illegible]

1. a) $\alpha)$ $y' = -4 \sin 4x$
 b) $y' = \frac{3(2 + \ln x)^2}{x} \times \frac{1}{x}$
 $= \frac{3(2 + \ln x)^2}{x^2}$
 c) $y' = \frac{x \cdot e^x - e^x \cdot 1}{x^2} = \frac{e^x(x-1)}{x^2}$
 d) $\alpha)$ $\frac{1}{4} \int \frac{4}{4x+1} dx = \frac{1}{4} \ln|4x+1| + C$
 b) $\frac{1}{20} \int 20x^3 e^{5x^4} dx = \frac{1}{20} e^{5x^4} + C$
 c) $\int 3x + 1 - \frac{2}{x} dx = \frac{3x^2}{2} + x - 2 \ln|x| + C$

2. a) $\int_{\frac{\pi}{4}}^{\pi} \cos x dx = [\sin x]_{\frac{\pi}{4}}^{\pi}$
 $= \sin \pi - \sin \frac{\pi}{4}$
 $= -\frac{1}{\sqrt{2}}$
 b) $y = \ln(3x+4) + 2x$ point $(-1, -2)$
 $y' = \frac{3}{3x+4} + 2$
 when $x = -1$ $y' = 5$ $\therefore m_{\text{normal}} = -\frac{1}{5}$
 $y - -2 = -\frac{1}{5}(x - -1)$
 $y = -\frac{1}{5}x - \frac{11}{5}$

c) Area = $\left| \int_{-1}^1 2e^{x+1} dx \right|$
 $= \left| [2e^{x+1}]_{-1}^1 \right|$
 $= 2e^2 - 2e^0$
 $= 2e^2 - 2$ or $2(e^2 - 1)$

d) $f'(x) = 2 \sec^2(2x)$
 $f(x) = \tan(2x) + C$ info $f\left(\frac{\pi}{4}\right) = 5$
 $\therefore 5 = \tan\left(2 \times \frac{\pi}{4}\right) + C$
 $5 = 1 + C$
 $C = 4$
 $\therefore f(x) = \tan 2x + 4$

3. a) i) $y = e^{3x}(x^2 - 5x + 3)$ product rule
 $y' = e^{3x} \cdot (2x - 5) + 3e^{3x} \cdot (x^2 - 5x + 3)$
 $= e^{3x}(2x - 5 + 3x^2 - 15x + 9)$
 $= e^{3x}(3x^2 - 13x + 4)$
 ii) stat pts set $y' = 0$
 $0 = e^{3x}(3x^2 - 13x + 4)$
 $0 = e^{3x}(3x - 1)(x - 4)$
 $x = \frac{1}{3}$ or $x = 4$
 $y = \frac{136}{9}$ $y = -e^{12}$

b) i) $\frac{d}{dx} (x+1)^3 + 2 = 3(x+1)^2$
 ii) $\frac{1}{3} \int \frac{3(x+1)^2}{(x+1)^3 + 2} dx = \frac{1}{3} \ln |(x+1)^3 + 2| + C$
 iii) LHS = $\int_0^{\sqrt{22}-1} \frac{(x+1)^2}{(x+1)^3 + 2} dx$
 $= \frac{1}{3} \left[\ln |(x+1)^3 + 2| \right]_0^{\sqrt{22}-1}$
 $= \frac{1}{3} (\ln 24 - \ln 3)$
 $= \frac{1}{3} \ln 8$
 $= \frac{1}{3} \ln 2^3$
 $= \ln 2 \quad \therefore b = 2$

4.a) Area = $\left| \int_1^{3/2} \frac{1}{x-2} + 1 dx \right|$
 $= \left| \left[\ln |x-2| + x \right]_1^{3/2} \right|$
 $= \left| \left(\ln \left| -\frac{1}{2} \right| + \frac{3}{2} \right) - \left(\ln |-1| + 1 \right) \right|$
 $= \left| \ln \frac{1}{2} + \frac{1}{2} \right|$
 $= \left| -\ln 2 + \frac{1}{2} \right|$
 $= \ln 2 - \frac{1}{2}$

b) Area = triangle - $\int_{\pi/6}^{\pi} \sin x dx$
 triangle = $\frac{1}{2} \times \left(\frac{3\pi}{2} - \frac{5\pi}{6} \right) \times \frac{1}{2} = \frac{\pi}{6} u^2$
 $\int_{\pi/6}^{\pi} \sin x dx = - \left[\cos x \right]_{\pi/6}^{\pi}$
 $= - \left(-1 - \frac{\sqrt{3}}{2} \right)$
 $= 1 + \frac{\sqrt{3}}{2} u^2$
 $\therefore \text{Area} = \frac{\pi}{6} - 1 + \frac{\sqrt{3}}{2} u^2$

c) $\int_1^4 3a^x dx = 14$
 $\int_1^4 3a^x dx = \int_1^5 3a^{x-1} dx$
 $\int_1^5 3a^{x-1} - 3 dx = 14 - 9$
 $= 5 u^2$